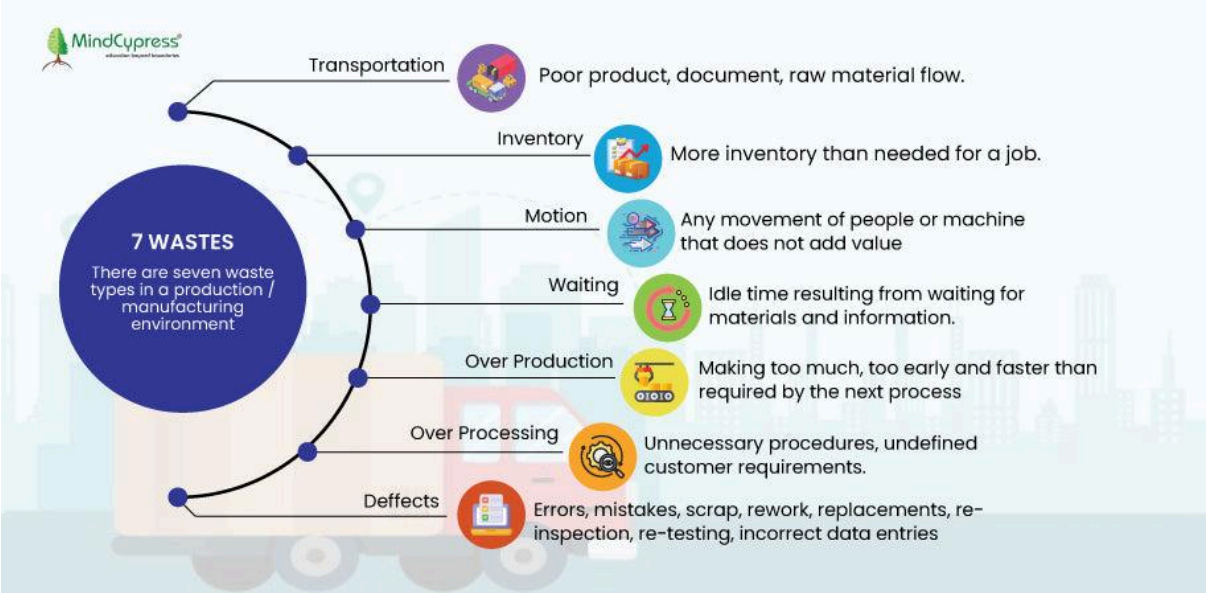
CHAPTER 6 — QUALITATIVE PROCESS ANALYSIS

(Finding what is wrong in a process and why, without numbers)



Where this chapter fits

- **Process Discovery** → how the process works
- **Qualitative Process Analysis** → what is wrong and where
- **Quantitative Analysis** → how bad it is in numbers

This chapter is **diagnostic**, not numerical.

---

1. Value-Added Analysis

(Is this step really worth doing?)

Core idea

Break the process into **small steps** and classify **each step** into one of three categories:

1. **Value-Adding (VA)**
2. **Business Value-Adding (BVA)**
3. **Non-Value-Adding (NVA)**

You do **not** classify the whole activity.
You classify **steps inside activities**.

1.1 Value-Adding (VA)

A step is **VA** if it **directly creates value for the customer**.

Ask **all three questions**:

- Would the customer be willing to pay for it?
- Does it help the customer achieve their goal?
- If removed, would the customer feel the service/product is worse?

If **yes**, it is VA.

Examples

- Delivering a product
- Assessing an admission application
- Confirming delivery date

👉 **Exam rule**
If customer value disappears → VA step

1.2 Business Value-Adding (BVA)

BVA steps **do not add value to the customer**, but are **necessary for the business**.

Ask:

- Is this required to earn revenue?
- Is it required for compliance or risk reduction?
- Would the business suffer if removed?

If yes → BVA.

Examples

- Credit check
- Invoice generation
- Degree verification
- Regulatory reporting

👉 **Key distinction**
Customer does not care, business must do it.

1.3 Non-Value-Adding (NVA)

Everything else.

Includes:

- Handovers
- Waiting
- Rework
- Corrections
- Delays

Examples

- Forwarding documents
- Waiting for approvals
- Resending information
- Receiving rejected items

👉 **Golden exam line**
NVA steps should be **eliminated or minimized**.

Example: Equipment Rental Process

- Fill request → VA
- Send request to clerk → NVA
- Select equipment → VA
- Forward request → NVA
- Produce PO → BVA

This is **classic exam material**.

---

2. Waste Analysis

(Why is the process slow, costly, or frustrating?)



Based on **Lean thinking**, waste = **anything that does not add value**.

2.1 MOVE (Unnecessary Transportation)

Moving information or material unnecessarily.

Examples:

- Printing → signing → scanning → emailing
- Sending documents back and forth

👉 Digital processes often hide MOVE waste.

2.2 MOTION

Unnecessary **movement of people or resources**.

Examples:

- Walking for signatures
- Switching systems repeatedly

Less common in IT, more in physical processes.

2.3 HOLD (Inventory / WIP)

Work that has started but not finished.

Examples:

- Pending applications
- Vehicles waiting for inspection
- Thousands of open cases

👉 WIP is **hidden waste**.

2.4 WAITING

Idle time:

- Task waiting for data
- Task waiting for resource
- Resource waiting for work

Examples:

- Waiting for committee meeting
- Waiting for missing documents

2.5 OVER-DO (Defects / Rework)

Redoing work due to errors.

Examples:

- Application correction
- Rechecking incomplete forms
- Reassessment due to missing data

2.6 OVER-PROCESSING

Doing **more work than necessary**.

Examples:

- Excessively detailed checks
- High accuracy where simple rejection is obvious

2.7 OVER-PRODUCTION

Doing work that is never used.

Examples:

- Quotes that never become orders
- Approved requests that get canceled
- Applications that are rejected anyway

👉 **Exam trick**
Most processes suffer from **multiple wastes simultaneously**.

---

3. Stakeholder Analysis

(Who complains about what?)

Purpose:

- Collect issues from **multiple perspectives**
- Avoid one-sided view

3.1 Stakeholder Categories

1. **Customers**
    - Slow service
    - Errors
    - Lack of transparency
2. **Process Participants**
    - Workload
    - Stress

    - Rework
3. **External Parties**
    - Unpredictable demand
    - Planning issues
4. **Process Owner**
    - Performance
    - Compliance
    - Bottlenecks
5. **Sponsors / Executives**
    - Strategic alignment
    - KPIs
    - Adaptability

👉 Same issue appears **differently** to different stakeholders.

---

4. Issue Register

(Turning complaints into structured data)

Purpose

- Organize issues
- Track impact
- Prioritize improvement

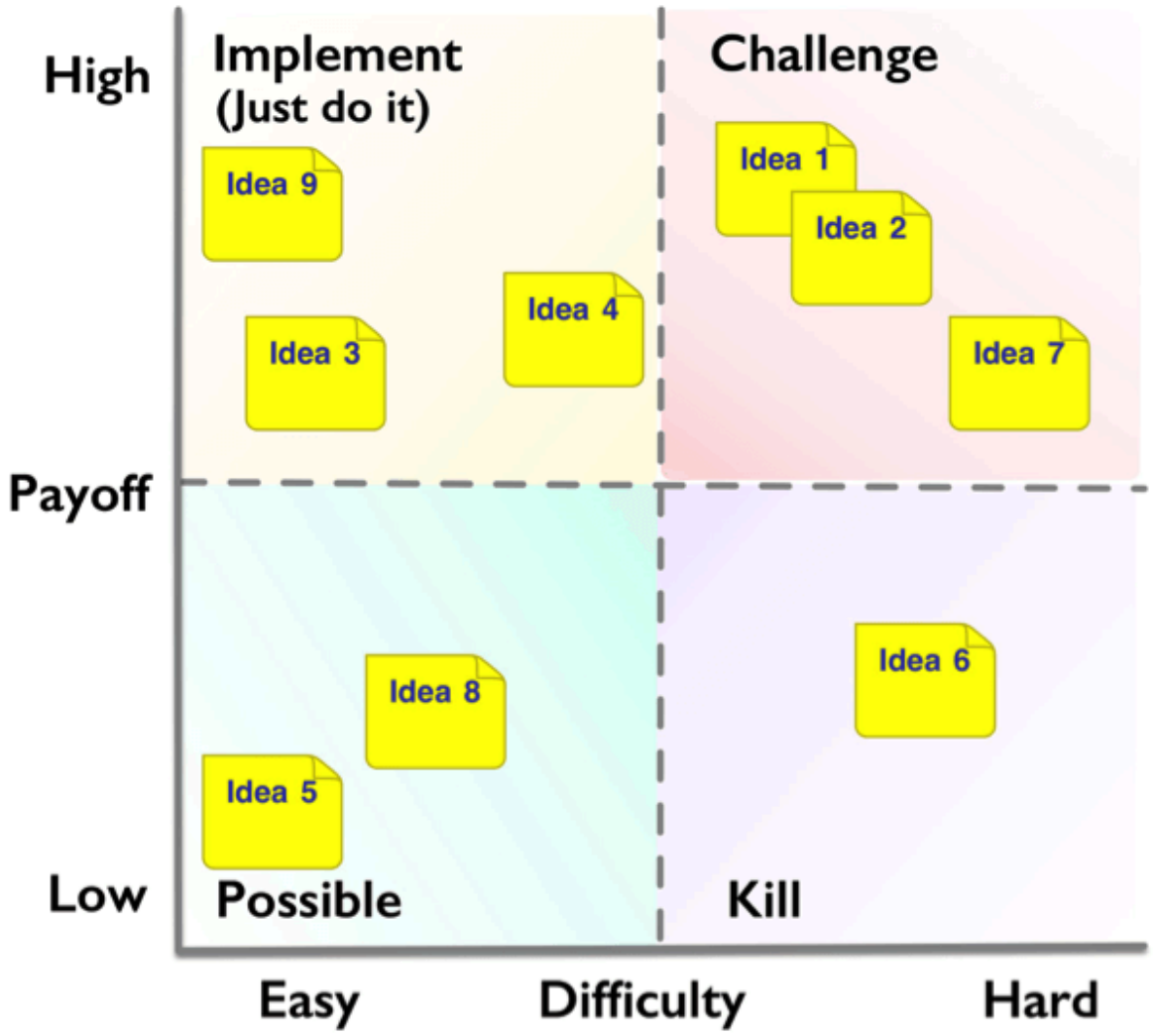
Typical Fields

- Issue ID
- Description
- Assumptions
- Impact (qualitative + quantitative)
- Possible solutions

👉 In exams, they often ask **“create an issue register”**.

---

5. Prioritization Techniques



5.1 Pareto Analysis

- 80 percent problems come from 20 percent causes
- Bar chart sorted by impact
- Cumulative curve on top

Use when:

- Many issues
- Limited improvement budget

5.2 PICK Chart

Two dimensions:

- Impact

- Effort

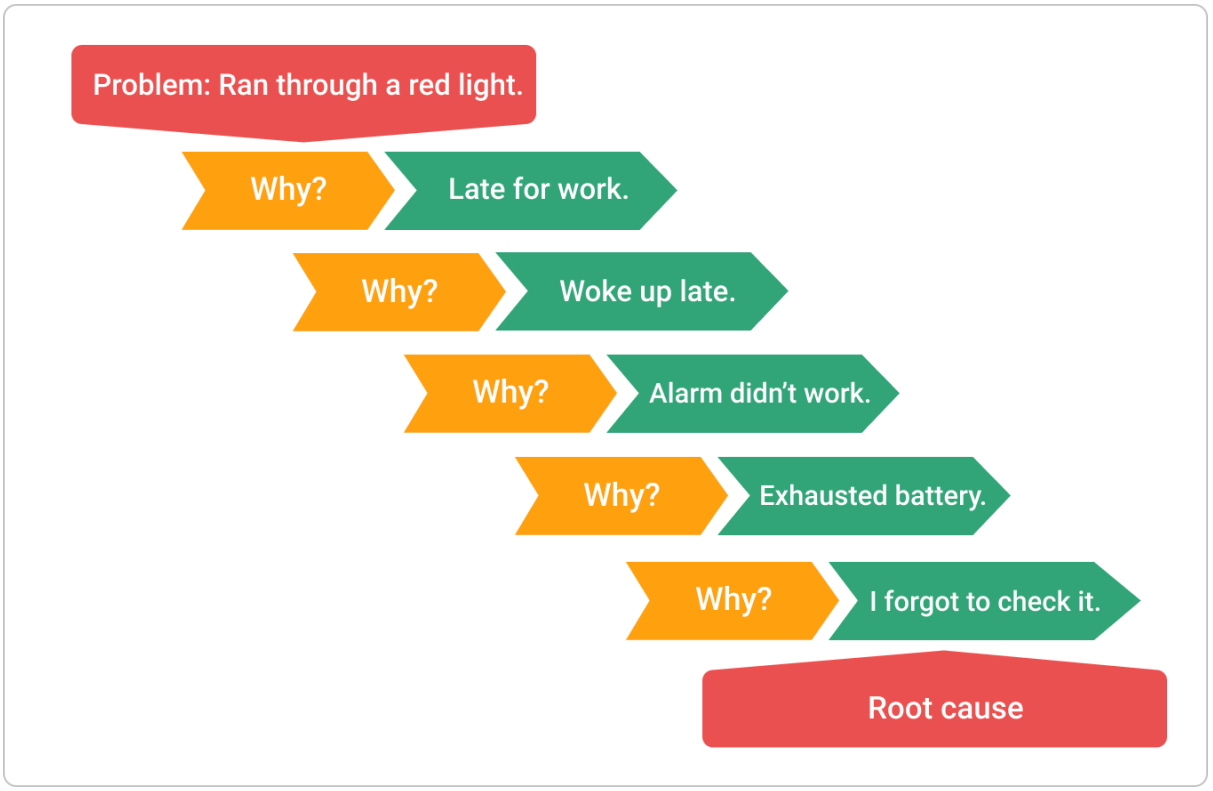
Quadrants:

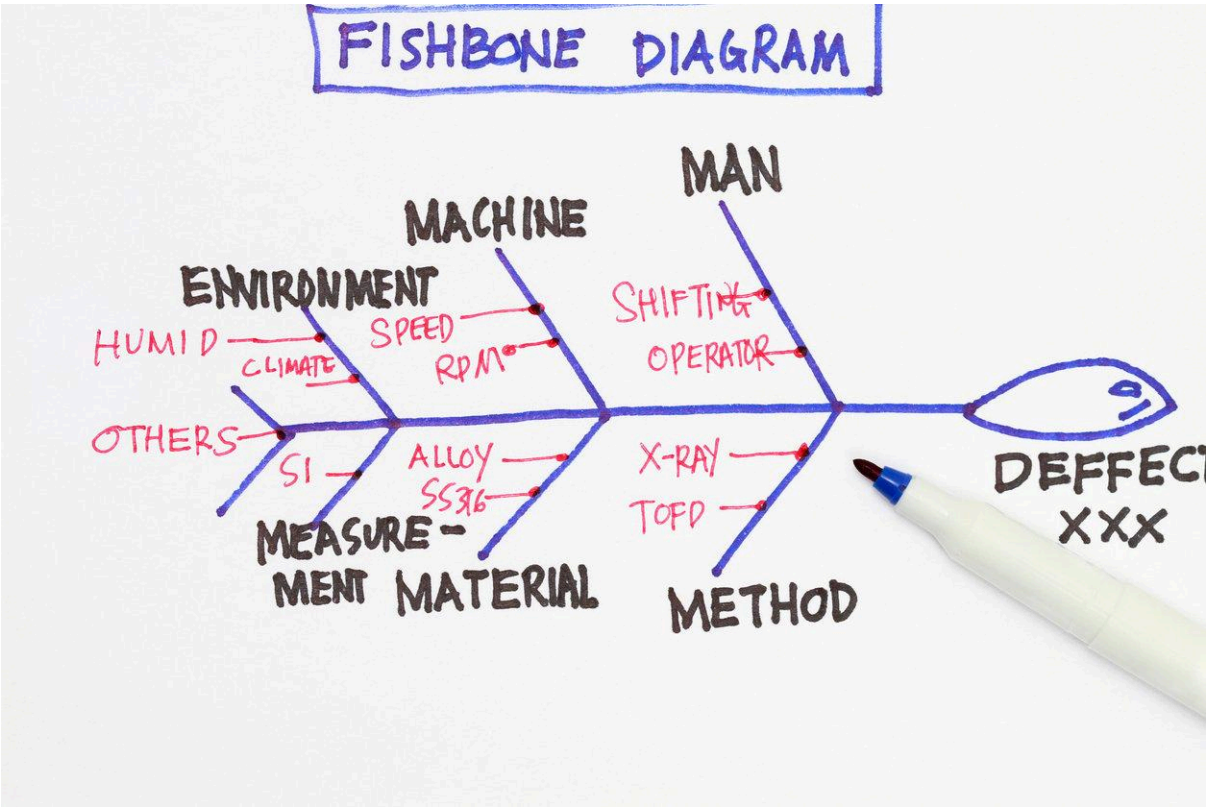
- **Implement** (high impact, low effort)
- **Possible**
- **Challenge**
- **Kill**

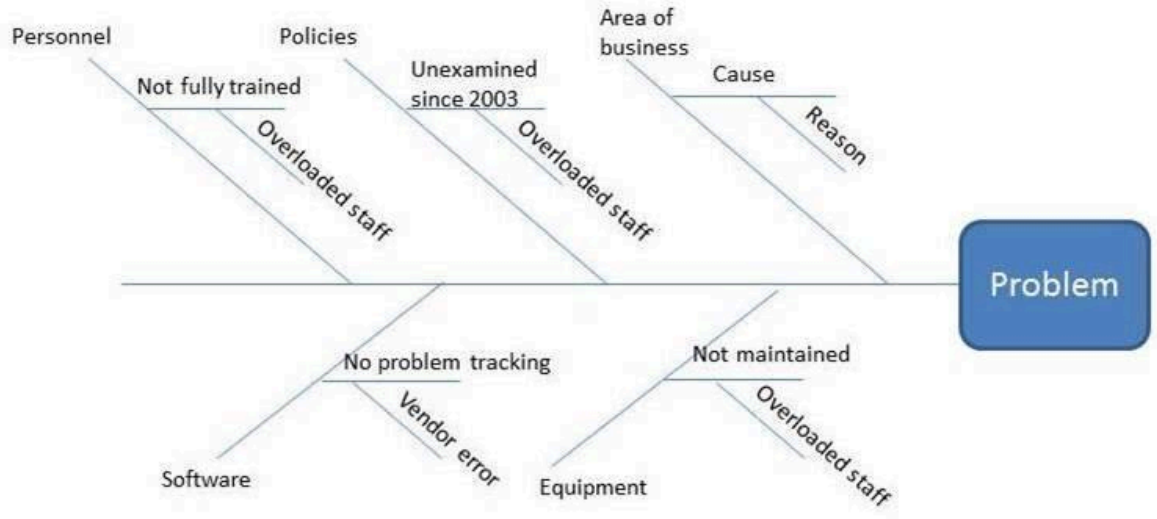
---

6. Root-Cause Analysis

(Why does the problem exist?)







6.1 Why-Why Analysis (Five Whys)

Keep asking **why** until you reach the real cause.

Example:

- Equipment kept too long → why?
- Fear of unavailability → why?
- Delivery takes too long → why?

- Approval slow → why?
- Sequential supplier contact

6.2 Cause-Effect Diagram (Fishbone)

Groups causes using **Six Ms**:

1. **Man** – skills, motivation
2. **Machine** – IT systems, tools
3. **Method** – process design
4. **Material** – data quality
5. **Measurement** – wrong metrics
6. **Milieu** – environment, external factors

👉 Excellent for **essay questions**.

FINAL RECAP (EXAM GOLD)

- VA → customer value
- BVA → business necessity
- NVA → waste
- Waste types → MOVE, MOTION, HOLD, WAIT, OVER-DO, OVER-PROCESS, OVER-PRODUCTION
- Stakeholders see **different problems**
- Issue register organizes issues
- Pareto & PICK prioritize
- Root-cause analysis explains **why**
